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DETECTING AI-GENERATED FAKE REVIEWS IN ONLINE PLATFORMS

DETEKCE FALEŠNÝCH RECENZÍ GENEROVANÝCH AI NA ONLINE PLATFORMÁCH

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Abstract

This thesis aims to implement a classifier for detecting fake reviews generated by artificial intelligence on shopping platforms, focusing on Slovak reviews. The thesis analyzes the impact of fake reviews on the reputation and sales of products and services. A synthetic dataset is created by scraping Slovak reviews from Booking.com, generating another set of reviews via a language model, and combining them. Then, the thesis presents a method for classifying these reviews. **[[It does this by this and that]].** **[[The results are this and that]].**

Abstrakt

Táto práca sa zameriava na implementáciu klasifikátora na detekciu falošných recenzií na shopping platformách, so zameraním na slovenské recenzie. Práca analyzuje reputačný a finančný dopad falošných recenzií na produkty a služby. Práca taktiež vytvára syntetický dataset zbieraním slovenských recenzií z Booking.com a ich kombináciou s recenziami vygenerovanými jazykovým modelom. **[[Robí to tak a tak.]]** **[[Výsledky sú to a to]].**

Keywords

Fake reviews, Machine learning, Artificial Intelligence, Text classification, Natural language processing, Python, Large language models

Klíčová slova

Falošné recenzie, Strojové učenie, Umelá inteligencia, Klasifikácia textu, Spracovanie prirodzeného jazyka, Python, Veľké jazykové modely

Reference

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Detecting AI-Generated Fake Reviews in Online Platforms

Declaration

I hereby declare that this Bachelor's thesis was prepared as an original work by the author under the supervision of Ing. Anton Firc, Ph.D. I have listed all the literary sources, publications and other sources, which were used during the preparation of this thesis. I have used large language models (ChatGPT, Gemini) for the language revision of the thesis. I have used Github Copilot to assist with the implementation part of the thesis.

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Igor Lacko
April 13, 2026

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Chapter 1

Introduction

Fake reviews have been a problem for online platforms since their inception. Reviews are often the first thing someone reads when considering buying a product or using a service. As such, they are one of the biggest influences in it's reputation, which in turn influences it's sales. Moreover, people may not even buy a quality product if it's reputation is damaged by fake reviews, or in a opposite example, may not buy a low quality product if a product is not artificially propped up. It's no wonder that people or companies often try to take advantage of this, whether it's for their advantage via faking positive reviews on their products or to the disadvantage of potential competitors via faking negative reviews on their products. The „traditional“ way of faking reviews in the past has been to manually write them, or pay for someone else to do it, which is time consuming and costly due to requiring manual human labor. However, with the rise of generative language models, such as *GPT*, and the ease of access to them, fake reviews can now be mass produced for cheap. The detection of fake reviews is a continuously developing field. This is due to multiple factors: human written fake reviews differ from AI generated ones, and even among the latter, as language models keep evolving, the methods for detecting their output have to as well. As humans have been shown not to be consistently able to correctly classify reviews as real or fake, as well as manual classification not being feasible from a time perspective, there are multiple benefits to automating this process. A common way to do this is by using machine learning (ML) methods, whether via more „traditional“ ML models such as *Naive Bayes*, *Logistic regression* or *Support vector machines (SVM)*, or more recently via utilizing language models. Commonly used features for these models are stylometric ones such as text length or writing style. However, these approaches, introduce another challenge, which is collecting relevant testing and training data. Existing datasets are few and far between, overly represent the hospitality domain, are mostly in English and often focus on manual spam instead of AI-generated reviews. This work aims to create a synthetic dataset in the Slovak language by combining existing reviews written in Slovak, using a generative model to generate the „fake half“ of the dataset, and train a classifier on the created dataset for the purpose of detecting reviews generated by language models. The classifier is an ensemble model which utilizes a pre-trained BERT-like model for the Slovak language in addition to classifiers who take in various stylometric features and other textual features, such as TF-IDF. The text of this thesis is structured as follows: **Chapter 2** describes the problem of fake reviews, the motivations behind them, and how can they affect consumers. **Chapter 3** then talks about the common features and techniques used in the domain of fake review detection up to this point, including both traditional machine learning methods and detection using transformers based language models. **Chapter 4** describes the dataset that

was used, how the real reviews were collected, and how it's synthetic half was generated. Chapter 5 describes a high level architecture of the proposed classifier and it's training algorithm. Chapter 6 describes the classifier implementation in more detail, including which models and features were used. Chapter 7 then discusses the results, and **Chapter 8** concludes the thesis.

Chapter 2

Fake reviews

Fake reviews pose a huge problem to both customers and sellers. They reduce the trust of customers in online platforms and thus have an inherently negative commercial impact on online available products.

2.1 The Dead Internet Theory

2.2 The influence of fake reviews

Fake reviews can have various negative effects. For example, a customer may not buy a product because of negative reviews associated with it. This is obviously unjust if there is a disinformation campaign against the product. On the other side, a product may get higher sales due to being artificially propped up by positive reviews, as a tactic of the seller to gain a market advantage over the competition. **Figure 1** shows a model of how different actors may interact with fake product reviews, including utilizing language models.

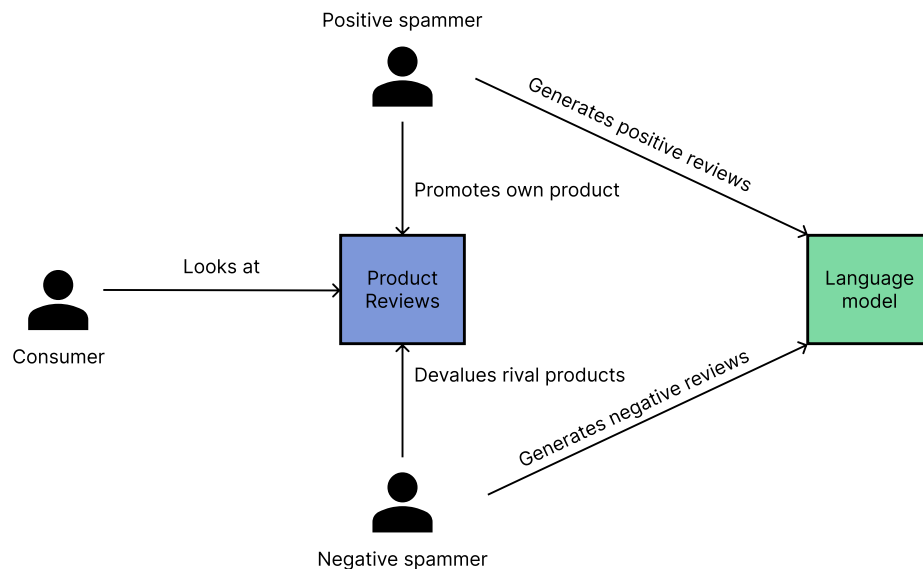


Figure 1: Spammers may promote or devalue products for market advantage. Consumers look at and decide based on them.

Influence on customer purchasing behavior

Perhaps the biggest factor in deciding whether to buy something is its reputation. That is why reviews exist, to present that information to people to better decide. The value of this is rooted in the presumed truthfulness of the reviews [15], which spammers¹ take advantage of. This doesn't only impact products ordered from the internet. [40] shows that around 93% of consumers say that online reviews influence their choice. More recently, [31] found that more than 70% of consumers read business reviews regularly when searching for local businesses and another 26% do so occasionally. Fake reviews can impact customer purchasing behavior in several ways. Firstly, a product with positive fake reviews can lead customers to buy it more, which can be harmful as fake reviews don't provide anything truthful regarding the value of the product. On the other side are negative reviews, which can cause consumers to distrust the product and also the seller [21]. Lastly, the knowledge of fake reviews existing itself can lead customers to distrust the review system itself and doubt all reviews [41], which can lower their purchasing intentions, indirectly harming the sales of online products or services.

Commercial impact of fake reviews and reviews in general

As the prior section implies, reviews can influence consumers purchasing behavior. This, in turn, can have real financial impacts on the reviewed products. Several studies show correlation between review ratings and sales. For example, [22] shows that a 1 star increase in Yelp rating leads to around a 9% increase in revenue, as an example of what may lead spammers to generate untruthful reviews. Considering only fake reviews, a 2021 study by the World Economic Forum [24] found that the impact of fake reviews on online spending is around 152 billion. They based this on the total influence of reviews, which, according to a study they cite is was 3.8 trillion in 2021 and their finding on average percentage of fake reviews on e-commerce sites, which they found to be 4%. This study also says that the eventual culling of fake reviews happens in general around after 100 days. This shows that fake reviews may even be a low-risk/high-reward situation for businesses who try to promote themselves throughout them, since the considerable gains outweigh the comparatively low risk. More recently, [1] gathered statistics concerning fake reviews for 2025. They estimate that fake reviews cost customers more than 700 billion in 2025, that around 30% of online reviews are considered fake, and that more than 80% of customers encounter them. Interestingly, they found that the number of fake reviews grows about 12% faster than the number of other reviews, which may indicate increased AI usage for fabricating fake reviews. Another interesting thing they found is that 46% of the reviews considered fake have been positive in sentiment (5 stars). This correlates with other studies, such as [15] which also attribute overly positive sentiment as a possible indication of fake reviews.

Review bombing

Review bombing is a phenomenon that has become more and more known to the public in the last years. Most people think of it in the context of entertainment media such as video games, movies, or TV shows [16] talks about review bombing in the context of the 2020 video game *The Last of Us: Part II* and its reviews on the review site *Metacritic*².

¹spammer — a person who posts fake reviews [28]

²<https://www.metacritic.com/>

They define review bombing as a coordinated action, where a lot of reviews are spammed on a product to alter its evaluation. They talk about review bombing as an *ideologically driven* action, as opposed to the traditional motivations behind fake reviews (e.g. market advantage). An interesting observation they found is that the reviews follow a bimodal distribution, with spikes at very low and very high ratings, which indicates that users may try to bomb a product „in counter“ to the opposite review bombing, such as users massively spamming positive reviews to balance out the negative ones. [35] found that a sign of review bombing can be a gap between user and critic ratings. This correlates with the aforementioned study, considering that *The Last of Us: Part II* won the 2020 Game of The Year award. Both they and [16] mention some of politics, race, gender, or sexual identity as frequently discussed topics in review bombing.

2.3 Generating fake reviews using AI

In the past, fabricating fake reviews was usually done by manually writing them or buying them. Studies such as [19] show that there exists a market for fake reviews. As the cited study says, a popular communication channel for procuring fake reviews were private Facebook groups. Nowadays, malicious actors have another method of obtaining fake reviews – using generative language models. There are many options here, such as using available APIs (for example OpenAI’s) for text generation, or just using a pre-trained generative model from the Hugging Face hub for review generation, or even just opening a language model web interface and asking to generate a review. [20] analyzed the frequency of AI-generated google reviews. They found that the rate nearly doubled from 2023 to 2024, and by 2024 almost one in five reviews on google is AI-generated. In addition to that, the quality of AI-generated reviews may be often higher than manually fabricated ones. [15] mentioned that humans have trouble distinguishing AI-generated reviews from real ones, often being no better than a random guesser (around 50%). Furthermore, concerning manually fabricated fake reviews, it mentions the inability of spammers to express true sentiment, which distinguishes their reviews from true ones. Large generative models whose training data includes reviews may not have this limitation. Another thing to consider is the ease of generating fake reviews. Manually writing fake reviews is time consuming, and the result is small in quantity. Just an online LLM can easily output 10 reviews in a single prompt. A python script controlling a loaded model for inference can output thousands in a tiny fraction of the time it would take to write them manually depending on the spammer’s hardware.

Chapter 3

Detecting AI generated reviews

Detection of AI-generated reviews is a field that is still very much in development. Despite the first research papers concerning fake review detection in general (e.g. including human-written fakes) dating back years, the number of studies concerning fake reviews has largely increased, and specifically concerning AI-generated reviews. Virtually all today's studies utilize some form of Machine Learning, whether it be traditional ML models or Deep Learning models.

3.1 Commonly used features in fake review detection

Stylometric features

Stylometry is the study of writing style. Several studies utilize it for feature extraction, since language models seem to have different writing style than humans. Some of the most common stylometric features are the frequency of using punctuation, average sentence or word length, n-gram features, or how often the review repeats words ([28], [17], [30]). Parts of speech can also be used as stylometric features. They can be used in various ways, such as the frequency of various POS in a given review [28], or the ratios of the POS tags among themselves [\[\[cite this\]\]](#).

Concerning unique words, one of the best known features concerning lexical diversity is the *Type-Token Ratio (TTR)* [23]. It is defined for a text T and individual tokens t as:

$$TTR(T) = \frac{|\{t \mid t \in T\}|}{|T|} \quad (3.1)$$

It's a number inside $< 0, 1 >$ indicating how lexically diverse the input text is. It could be useful to detect a model using a limited set of words.

Hapax-Legomenon rate is a feature that could also evaluate the lexical diversity of a text [30]. It is defined for a text T and tokens t as:

$$HLR(T) = \frac{|\{t \mid count(t, T) = 1\}|}{|T|} \quad (3.2)$$

Interestingly enough, it was one of the 4 most important features in [30]. It could detect the usage of uncommon words, but also for example a model with parameters that punish repetition.

Another useful metric which measures the simplicity of a input text is the *Automated Readability Index (ARI)*. Parts of the following paragraph about it are paraphrased from [15]. For a given text, ARI is defined as:

$$ARI(text) = 4.71 \frac{|chars(text)|}{|words(text)|} + 0.5 \frac{|words(text)|}{|sentences(text)|} - 21.43 \quad (3.3)$$

ARI is a readability metric which indicates the US education level needed to understand the text. A score around 3 means that a US third grader can understand it, etc. It is similar to TTR in that a higher number usually indicates a more complex text.

N-grams are sequences of elements of length N. „Elements“ most usually refers to words here, but there can be character or token based N-grams [2]. Unigrams have 1 element, bigrams have 2, and so on. For example, if we generalize elements to words, and ask which bigrams are in the sentence “*Cats are very cute*”, they would be:

$$\{(Cats, are), (are, very), (very, cute)\}$$

N-grams can be useful features for finding out which phrases are the most common in the input text. For example, if a language model tends to repeat the same phrases often, the n-grams corresponding to these phrases will appear more often than other ones. This could be especially useful in something as specific as fake review detection, where the models used to generate spam could tend to repeat the same positive or negative phrases.

TF-IDF (Term Frequency-Inverse Document Frequency) is a feature extraction method. It most usually works with individual words, but it can be character level or n-gram level. The individual elements are called *terms*. Parts of the following paragraph are taken from [10]. To compute the vector representation of the input text, TF-IDF uses two vectors. For a term t and a dataset entry E , the TF vector is defined as:

$$TF(t, E) = \frac{count(t, E)}{|E|} \quad (3.4)$$

In simple terms, it’s a score which is indicating how relevant t is in E . The more often it appears, the higher the TF score will be.

The second vector, IDF is composed of elements defined for a term t as:

$$IDF(t) = \log\left(\frac{|D|}{|\{E \in D \mid t \in E\}|}\right) \quad (3.5)$$

This vector assigns a higher score to terms which appear in less documents, hence the *inverse*. As opposed to TF, this vector is mutual for the entire dataset, and as such does not need to be computed for every text inside.

The resulting vectorized text, TF-IDF is then defined as:

$$TF-IDF(t, E) = TF(t, E) \cdot IDF(t) \quad (3.6)$$

meaning that the highest score terms are the ones, which are relevant to that specific entry but don’t appear too often in other entries.

Semantic features

Semantic features focus on the meaning of the text instead of its structure. This can allow them to model relationships between words that aren't seen just from the structure, or extract deeper meaning from the review.

A common semantic feature is the *sentiment* of the review. Per multiple studies ([38], [18], [15]) polarized sentiment is prevalent in fake reviews. That means that fake reviews tend to be very positive or very negative, but very few „in the middle“. Per [18] however, positive fake reviews outnumber negative fake reviews (which might be expected, since promoting own product seems like it would have more influence than defaming the competition). Review sentiment can be extracted in various ways, such as lexicon based tools [38] or by using a sentiment classifier.

Word embeddings are a widely used method to capture semantic features. The underlying features are more of a „black box“ to us, since we don't know what the individual dimensions in the embedding vector space represent, but they are a very effective method to turn text into numbers and capture more than its structure. Word embeddings are primarily used as features with deep learning models (various neural network architectures, Transformer-based models) but can be used as features with other classifiers as well. For example, [27] used a BERT model to extract contextualized word embeddings, and then trained multiple classifiers on these embeddings, such as SVM, Random Forest, and more.

Review and product metadata

Review metadata are features about the review, but not about its text. They go hand in hand with metadata about the product. Examples of this include review rating, product ID, or date and time posted ([28], [17]). One of the aforementioned studies ([15]) which talks about the polarized sentiment of fake reviews also found that fake reviews tend to follow a bimodal rating distribution, meaning they have sharp peaks at the lowest and highest rating. Other studies have found that other ratings of the product could also be a useful feature, with [37] finding that the number of 1 star ratings on a product was one of the most important features in their Random Forest model.

Reviewer centric features

Reviewer centric features are characteristics of the reviewer's profile. [28] lists some of these features. They include features such as repeated reviews across products reviewed, sentiment of the reviews (for a concrete reviewer), or burstiness (tendency to post a lot of reviews in a short time span). Another survey [17] also lists the absence of profile data, or review time span. These features can greatly improve the results of the detection [15, 28, 17] **[verify]**, however they are difficult to obtain.

3.2 Traditional Machine Learning models

SVM

The information in this section is taken from [36]. *Support Vector Machines (SVM)* are models, which try to separate the input features by a hyperplane. Basically, the model tries to find a hyperplane from the input features x :

$$\vec{w} \cdot \vec{x} + b = 0 \quad (3.7)$$

When using only two classes, the model then tries to map a new point to one of them using this hyperplane depending on which side of the hyperplane the new, unlabeled point lands on. If we express the classes as $\{-1, 1\}$, then the classification function f can be expressed as:

$$f(x) = \begin{cases} -1 & \text{if } \vec{w} \cdot \vec{x} + b < 0 \\ 1 & \text{if } \vec{w} \cdot \vec{x} + b \geq 0 \end{cases} \quad (3.8)$$

The concrete hyperplane the model tries to find, is the one with the largest *margin*, which is the minimal distance of the hyperplane to a point from one of the defined classes. The idea behind this is that if the margin is high, then the classes are better separated, meaning it's harder for the model to label predictions incorrectly. However, since the points may not be always separable by a hyperplane in the number of dimensions of the feature space (take for example a 2D feature space with one class concentrated in a circle, and another class in a larger circle around it). The *kernel trick* is a method of moving the feature space to a higher dimension where the classes are separable, and constructing the hyperplane there. This is done by using a *kernel function*, which computes the hyperplane in a higher dimensional space, and it does so without actually moving the samples to that space, only computing the relationship between the samples „as if they were“ in the higher dimensional space. **Figure 2** shows an example of this considering 2D and 3D spaces. ¹

¹Picture taken from <https://gregorygundersen.com/blog/2019/12/10/kernel-trick/>

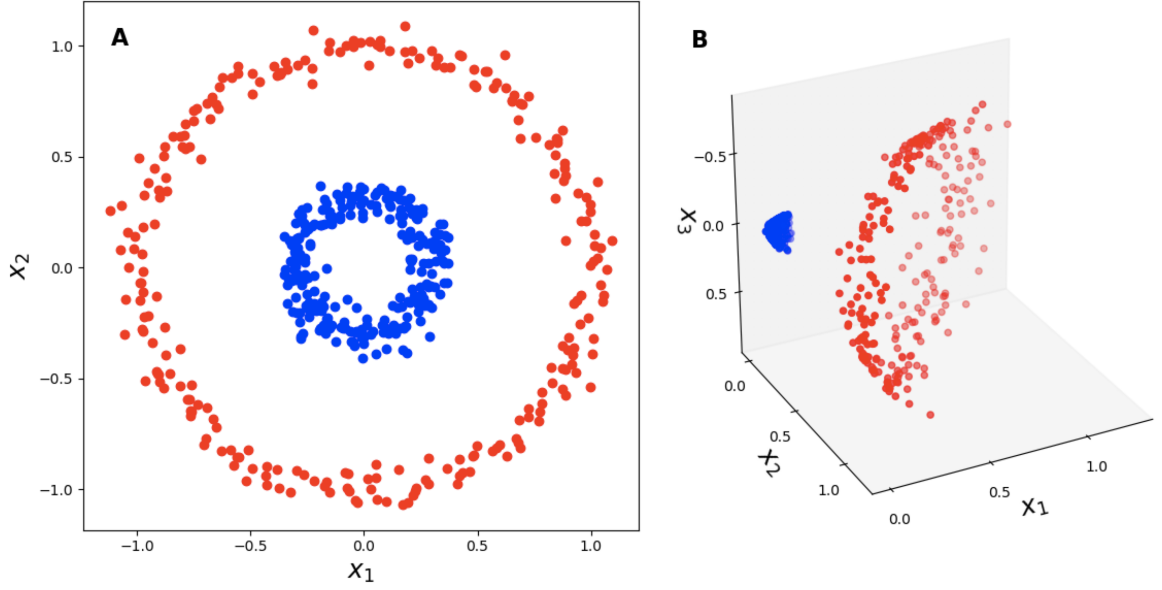


Figure 2: Input features in 2D where they aren't separable by a 2D hyperplane (a line) and in 3D, where they are separable by a 3D hyperplane (a plane).

An example is the *polynomial kernel*, defined as:

$$K(\vec{x}, \vec{x}_i) = ((\vec{x}^T \vec{x}_i) + \eta)^d \quad (3.9)$$

Where $\vec{x}$ is a unlabeled example the model is trying to classify, $\vec{x}_i$ is a training example, η is a hyperparameter and d is the number of dimensions of the higher dimensional space.

Decision Trees and Random Forests

The information in this section is taken from [25]. Decision trees are models, which continuously split the data based on values of different features until a stopping criterion is reached. This builds a tree like structure, where the root node represents the entire dataset, and the leaf nodes are individual class labels. An example of a decision tree algorithm is ID3 (*Iterative Dichotomiser 3*), which chooses the feature to split on based on information gain. Also, one of the biggest advantages of decision trees are that they are easy to visualize and interpret, for example when evaluating feature importance, as opposed to for example Deep Learning models. *Random Forests* are models built from several decision trees. These trees have random subsets from the dataset and a random subset of features. This can improve vulnerabilities that normal decision trees are prone to, such as overfitting. Figure 3 shows a high level overview of how random forests work. ²

²Picture taken from <https://dida.do/what-is-random-forest>.

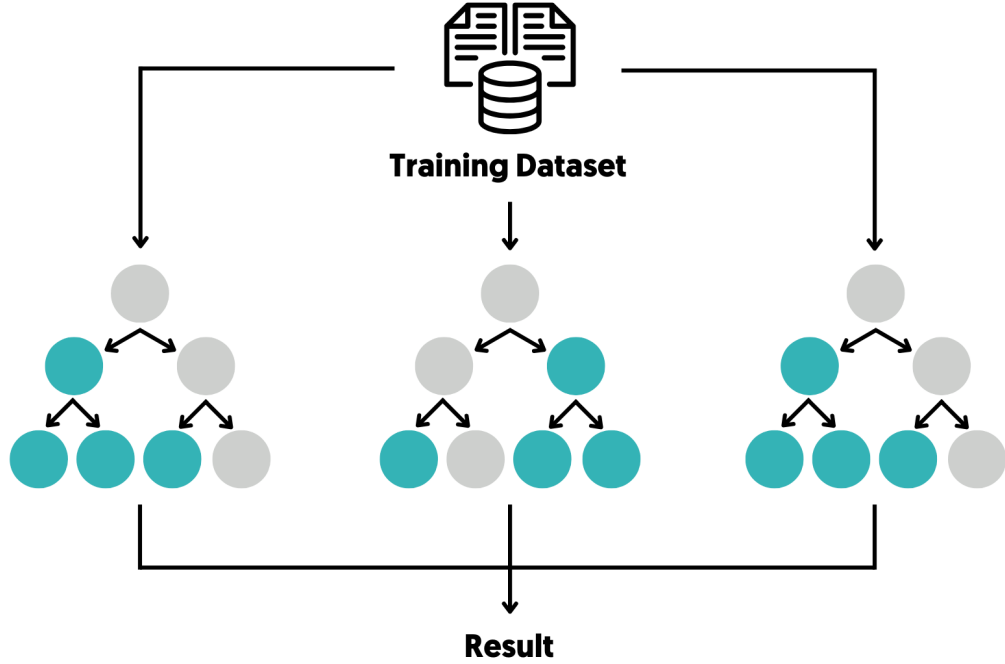


Figure 3: High level overview of Random Forests - each tree takes a random subset of the dataset. The predictions of the trees are averaged into the final prediction.

Naive Bayes

The *Naive Bayes* classifiers are a family of models based on the *Bayes Theorem*, which is a way of calculating a conditional probability of a event given another event. It uses the reverse conditional probability and the independent probabilities of the two events. It is defined as [7]:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)} \quad (3.10)$$

When using this for classification tasks the model tries to calculate the probability of a given class, given the input data (features) [7]. For example, for AI generated review detection, for the review „*I hate this hotel!*“ we are trying to find $P(\text{AI-generated} | \text{I hate this hotel})$, meaning the probability the review is AI-generated, given that it is „*I hate this hotel!*“. The model makes the *naive* assumption that all input variables are independent of each other. For text classification, if we take words/unigrams as features, this assumption means that the word orders are independent of each other [34]. An implication of this is that the model only considers word frequencies, and the review „*The hotel was good, but the food was terrible!*“ will be always classified the same as „*The hotel was terrible, but the food was good!*“ despite one probably having a higher rating than the other. However, this massively simplifies the computation, as it means that the overall probability can be split into a product of probabilities of each feature [7]. In this example, that would mean that:

$$P(\text{AI-generated} | \text{I hate this hotel!}) = P(\text{AI-generated} | I) * ... P(\text{AI-generated} | \text{hotel}) \quad (3.11)$$

While the classifier has multiple versions, the one which considers a document as a sequence of words and uses word frequencies is called *Multinomial Naive Bayes* [34].

Logistic Regression

Despite its name, *Logistic Regression* is a model primarily used for classification. As opposed to normal *Linear Regression*, which predicts a *continuous* output, it predicts a *categorical* output, that being the probability (a number in $< 0, 1 >$) of the output falling into a category [32, 13]. Another misleading thing about its name is that it is still a *linear* model. However, since the output of a linear model is continuous, and belongs in $(-\infty, \infty)$, while probabilities belong in $< 0, 1 >$ [13], the model isn't linear between the input features and the output, but instead between the input features and the *logits* of the output, which in essence means that it predicts $\text{logit}(Y)$ instead of Y from X [32, 13]. The *logit* of a variable Y is defined as:

$$\text{logit}(Y) = \ln(\text{odds}(Y)) \quad (3.12)$$

where odds are the probability of Y happening, e.g. $p(Y)$ relative to the probability of Y not happening, e.g. $1 - p(Y)$. Then the equation can be written as:

$$\text{logit}(Y) = \ln\left(\frac{p(Y)}{1 - p(Y)}\right) \quad (3.13)$$

and the model tries to equate this to a linear model, meaning it tries to solve:

$$\ln\left(\frac{p(Y)}{1 - p(Y)}\right) = \alpha + \beta X \quad (3.14)$$

Which, after exponentiation and simplifying, results in the following [32, 13]:

$$p(Y|X = x) = \frac{e^{\alpha + \beta X}}{1 + e^{\alpha + \beta X}} \quad (3.15)$$

This function always outputs a number between 0 and 1. Classification done by a threshold, usually 0.5 [13]. For example, if the probability of a review being fake exceeds 0.5, we classify it as fake, otherwise, we classify it as authentic.

KNN

The information from this subsection is taken from [9]. The *K-Nearest-Neighbors* classifier is very simplistic in its core idea. It classifies a new sample based on its closest neighbors among the training samples, and assigns it to their class. The variants of the classifier differ in three things:

1. The K in KNN, e.g. the number of neighbors used
2. The distance metric used to compute the nearest neighbors
3. The voting function, which determines the class of the new sample using the nearest neighbors

Common distance metrics include variants of the *Minkowski* distance, which for N features, that is $N - \text{dimensional}$ spaces is defined as:

$$d(y, x_i) = \left(\sum_{n=1}^N |y_n - x_{in}|^p \right)^{\frac{1}{p}} \quad (3.16)$$

And is most often used with $p = 1$, $p = 2$ or $p = \infty$ which are the Manhattan, Euclidean and Chebyshev distances respectively. For the voting function, a common one is to use a weighted one where the closest neighbors' votes weigh more than other.

3.3 Deep learning models

Neural networks

Transformers based language models

Modern large language models are based on the transformers architecture, which was introduced in the 2017 paper *Attention is all you need* [39]. What makes these models so special is the *self-attention* mechanism, which allows a model to add context to word embeddings based on the words around it [3], which solves the problem of the embeddings being static for each word. The original architecture is composed of encoder and decoder parts, where the encoder is responsible for contextualizing the input embedding and the decoder takes it as an input and generates the next word [39]. Both the Encoder and Decoder are a stack of identical layers, where each Encoder layer is composed of self-attention followed by a feed-forward neural network, while the Decoder is similar, but it masks future positions in the self-attention part to only make the new word depend on past ones, and it adds a *Encoder-Decoder attention* layer into the middle, which performs self-attention on the output of the final Encoder [39, 3]. However, most modern models only have one of these components. *Encoder* models, with only the Encoder component are often used for tasks such as classification and sentiment analysis, since they can create contextualized word embeddings [11]. The best-known model in this lineage is *BERT*, introduced in 2018 [11]. *Decoder models*, on the other hand, are used for generative tasks. The best known Decoder only model is the *GPT* model [33], also introduced in 2018, which is the model whose newer version is behind the *Chat-GPT* web chat interface. While Encoder model usage, such as in [27] would be expected for fake review classification, [15] also used *GPT-Neo*, a Decoder model to classify fake reviews with good results.

3.4 Ensemble models

Chapter 4

The Booking.com synthetic review dataset

The dataset used in this thesis was collected using reviews of Slovak hotels on Booking.com. From the scraped reviews, two variants were created. One where each review was used as „unified“ content (meaning the positive and negative content were concatenated) and one where they were used as separate samples.

4.1 Choice of the data source

Part of the reason why Booking.com was chosen is due to inspiration by previous datasets used in fake review detection. The hospitality sector dominates the percentage of the used datasets, per [28] and [17]. Figure 2, taken from [17] and recreated into a SVG shows the distribution of the datasets used from 2021 to 2023:

Domains in fake review detection datasets from 2021 to 2023

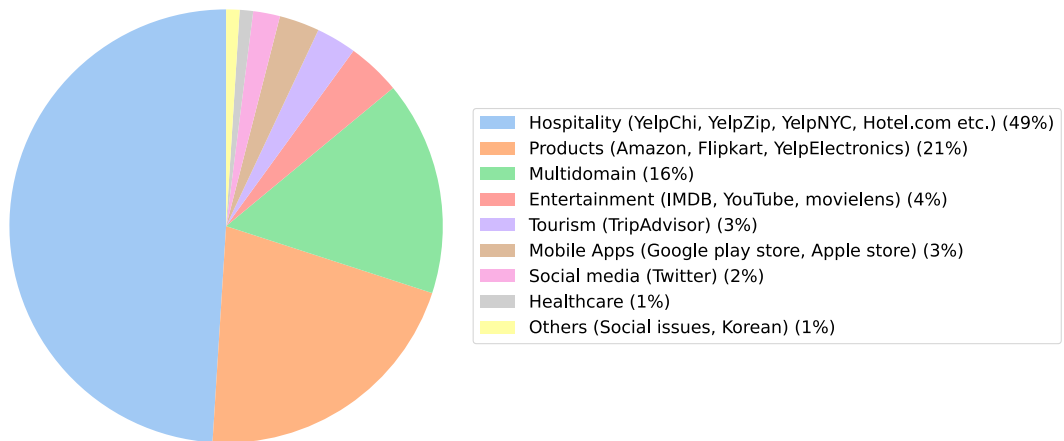


Figure 4: Distribution of dataset domains used in fake review detection from 2021 to 2023

Booking.com represents the largest source of reviews of hotels all over Slovakia, with options to filter by language to use only reviews in Slovak. The website can be easily parsed, and the reviews are structured into 2 parts (the „good part“, or „what did you like?“, and the

„bad part“, or „what did you dislike?“) which is more complicated than 1 review = 1 text, but allows for multiple variants (use a review as one sample or as 2 samples).

4.2 Collecting data

The process of scraping

A total of 10064 reviews were scraped from the Booking.com website. The scraper always navigated to the website and searched for hotels in Slovakia. Then it would go hotel by hotel, filter the reviews by language to Slovak, and scrape each review from a hotel which it didn't scrape already (known by URL). The scraper, which was a headless browser implemented in Python using Playwright would extract the hotel name (the individual CSV files were saved by their hotel name for filtering already scraped hotels), the positive content, negative content and rating and save them into a CSV.

Scraping permission and legal aspects of the used data

Since web scraping is a legally gray area, and Booking.com's terms and conditions [5] disallow any sort of monitoring, scraping and crawling, permission was asked from their data privacy office. They granted it under the conditions that the collected data is anonymized, the review content is not referred to, and the data is not shared. Per their request, the dataset does not contain any personal data, and it's content is not published other than aggregate statistics.

4.3 Dataset characteristics

This section describes the non synthetic part of the dataset, it's characteristics and statistics. The dataset contains 10064 rows, each of which is can be defined as a quadruple of ($rating, content_{good}, content_{bad}, content_{unified}$) where the last is just the two content parts concatenated with a whitespace between them if both are present (else it's identical to the present part). **Table 1** describes how many of the scraped reviews contain only positive and negative content, and how many contain both.

Content type	Count
Total reviews	10064
Only positive content	2587
Only negative content	370
Both positive and negative content	7107

Table 1: Statistic of content type presence in the reviews.

From the table we can see that about three quarters of the dataset have both negative and positive content. Out of those that don't, there are almost 7 times as many reviews with only positive content than reviews with only negative content. From this, it comes as no surprise that the dataset heavily skews towards positive ratings. Booking.com uses ratings in the interval $< 1, 10 >$ and **figure 3** shows that the mean rating is 8.24, and if we go by discrete bins, reviews with a rating of 10 are by far the most common. This might be due to Booking.com pushing highly rated hotels up as default results in a location.

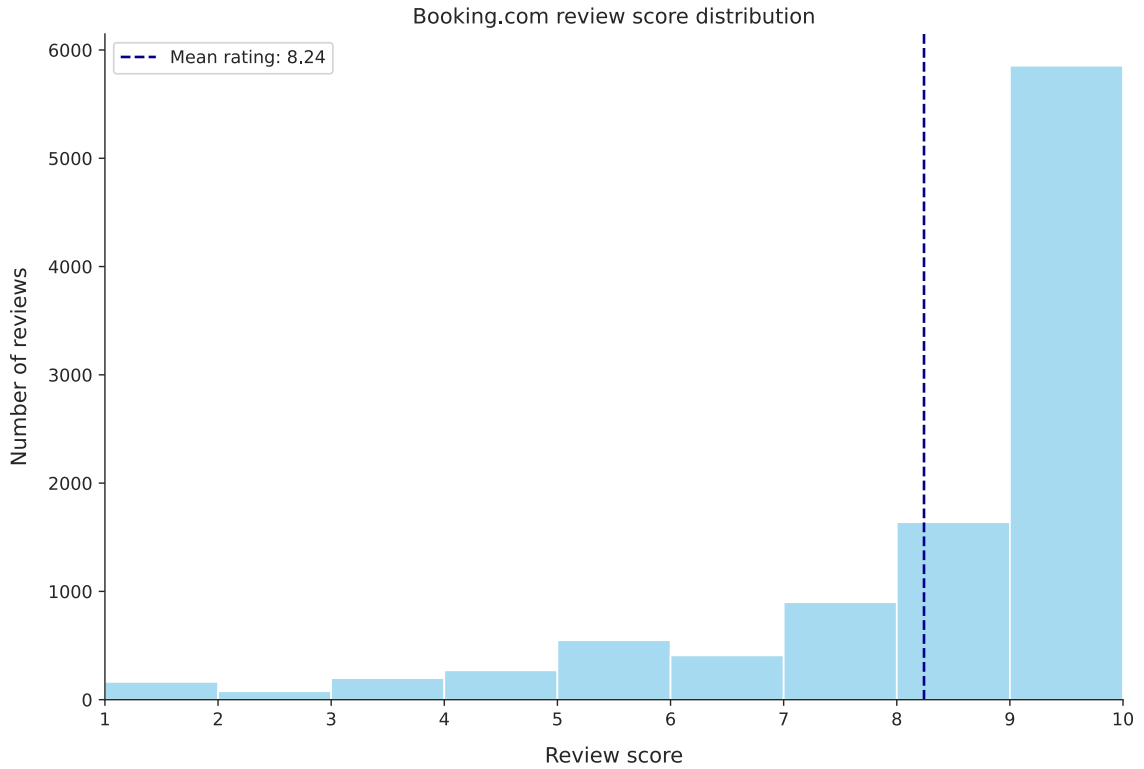


Figure 5: Distribution of ratings in the dataset.

When it comes to the text length, [figure 4](#) and [table 2](#) show that both the positive and negative parts of the reviews are mostly around 110 characters (in median), with long outliers skewing the mean to the right, but both parts don’t significantly differ from each other.

Content type	Average length	Median length	Max length	Min length
Positive	148.65	117.00	1988.0	1.0
Negative	159.18	106.00	1998.0	1.0
Total	262.15	195.00	2810	96

Table 2: Statistics of review lengths for different content types.

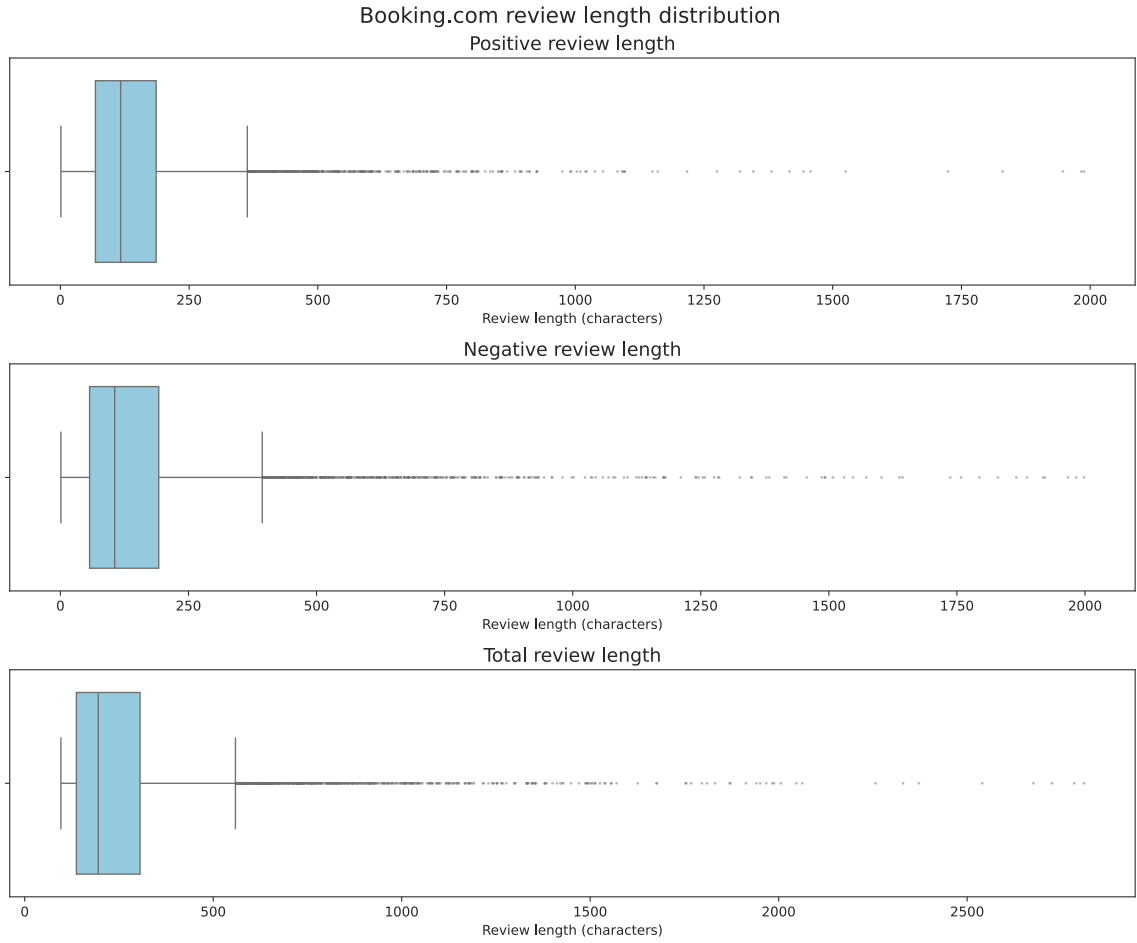


Figure 6: Distribution of review lengths in the dataset.

4.4 Generating synthetic data

Datasets used in the past have been constructed by methods such as crowdsourcing [28] or some other form of asking people to manually fabricate fake reviews. Another commonly used method was by using a filtering algorithm. One of the oldest datasets for fake review detection [29] collected hotel reviews from Yelp, which has its own filtering algorithm but doesn't delete reviews labeled as spam, but puts them in a separate list. However, the first method makes it difficult to gather large amounts of data, while the second labels reviews based on a non transparent algorithm. Another option is to create a new, synthetic dataset based on the „real“ one which contains scraped reviews. This involves using a generative language model to generate reviews and annotating them as fake. [15] which used GPT-3 and [26] which used various models have found success with this method. Its benefit is that it allows the generation of high quality annotated data very quickly, however we still need the „real“ part of the dataset.

Model comparison

For generating synthetic reviews, 3 models were evaluated. Two of them were instruct models. The first was a pre-quantized (into 4bit by Unsloth) 27b version of the Gemma 3 model

¹ by Google, and the second was a 32b version of the Qwen 3 model ² by Qwen. Both of those are known for their multilingual capabilities. The third model was the mistral-sk-7b model ³, which is a fine-tuned mistral model for the Slovak language. This model is a base model. Each of these models was used to generate 200 samples from the unified variant of the dataset, and 200 samples from the separate. Then the generated text quality was compared with 2 methods:

1. Perplexity: For each of the generated sample „datasets“, perplexity was computed on the whole set using an independent model.
2. Survey: A Google Forms survey was created, asking humans to rank the models by how „natural“ the text is.

Model prompts

The generative models were always prompted with example reviews from the dataset. To make the synthetic data follow the sentiment distribution of the real dataset, the examples were sampled as follows. One review was always randomly sampled from the entire dataset. Then:

1. For the separate variant: 5 examples were randomly chosen from the part of the dataset that matched it’s content type (positive /negative). The content type was stored alongside the new rating for later stratification into training/validation/testing sets.
2. For the unified variant: 5 examples were randomly chosen from the part of the dataset whose rating was in $< \max(\text{rating}_{\text{example}} - 1, 1), \min(\text{rating}_{\text{example}} + 1, 10) >$. The rating of the example was also stored alongside the new review.

The prompts itself however differed between models. Since the mistral model is not a instruct model, which means it’s not tuned to follow instructions, we couldn’t use a prompt like „Vygeneruj recenziu hotela po slovensky...“. Instead, the model was always left to fill in the last example. The generic prompt for this model then looked like:

```
Hotelová recenzia 1: <Example 1>
Hotelová recenzia 2: <Example 2>
...
Hotelová recenzia 6:
```

On the other hand, the instruct models allowed bigger flexibility with the prompt. Based on the sampled content type of rating, the prompt was always a bit different. The models followed a label instruction, which described what should the review’s sentiment look like. This is showcased in tables 3 and 4 for the unified and separate variant.

¹<https://huggingface.co/unsloth/gemma-3-27b-it-bnb-4bit>

²<https://huggingface.co/Qwen/Qwen3-32B>

³<https://huggingface.co/slovak-nlp/mistral-sk-7b>

Rating	Label instruction
≥ 8	veľmi spokojná
≥ 6	spokojná
≥ 3	nespokojná
< 3	veľmi nespokojná

Table 3: The label instructions based on the sampled rating.

Content type	Label instruction
$content_{good}$	o tom čo sa hosťom páčilo
$content_{bad}$	o tom čo sa hosťom nepáčilo

Table 4: The label instructions based on the sampled content type.

The models also produced much longer reviews than the given examples. To mitigate this, the min review word count, max review word count and average review word count were computed from the examples and included in the prompt, as:

```
f"Dĺžka má byť približne {target_words} slov "  
f"(rozsah {min_words}-{max_words} slov). "  
f"Nepíš viac ako {max_words} slov.
```

Model evaluation

Perplexity

Perplexity (*PPL*) is a generative LLM evaluation metric, which describes how much a model is confused („perplexed“) by a text, in the sense that it outputs how many tokens was it choosing from on average [6], which means that lower perplexity usually indicates more coherent text. It is computed as ([12]):

$$PPL(X) = \exp\left(-\frac{1}{t} \sum_{i=1}^t \log p_{\theta}(x_i|x_{<i})\right) \quad (4.1)$$

Where X is the text to compute perplexity for (in this case, one review), t is the length of the review in tokens, and $p_{\theta}(x_i|x_{<i})$ denotes the probability that a the i th token would follow the preceding tokens.

The perplexity was computed for each text in the sample datasets, summed up and then averaged. To avoid a situation where a model would assign itself a lower perplexity score, a independent model was used to compute this. The model used was the EuroLLM-22B-Instruct-2512 model ⁴, which has Slovak in the list of languages it understands. The results are visible in [table 5](#).

⁴<https://huggingface.co/utter-project/EuroLLM-22B-Instruct-2512>

	Model	Perplexity	Variant
0	Gemma	7.107292	Unified
1	Qwen	17.466667	Unified
2	Mistral	17.810422	Unified
3	Gemma	14.902512	Separate
4	Qwen	73.536782	Separate
5	Mistral	47.902919	Separate

Table 5: Perplexity scores for the individual models for each variant.

For both variants, the Gemma model has the best score. The other two models are close in the unified variant, but the mistral model is a distant second in the separate one.

Google Forms survey

The other technique used to evaluate the generated text was to let human evaluators read it and rank the models. A Google Forms survey was created with 10 questions. Each question contained one sample from each model, where 5 of the questions contained samples from the unified variant and 5 from the separate. The evaluators were asked to rank the examples from most to least natural sounding, e.g. to rank the one where they would most likely believe that it was human written at the top, and the opposite at the bottom. Each model has assigned a point for a first place vote, half of a point for a second place vote, and zero points for a third place vote. So the total score the model received was:

$$Score = votes_{first} + \frac{votes_{second}}{2} \quad (4.2)$$

The results are visible in [table 6](#).

	First place votes	Second place votes	Third place votes	Total score
Gemma	39	25	46	51.5
Mistral	49	36	25	67
Qwen	22	49	39	46.5

Table 6: Results of the Google Forms survey.

As opposed to perplexity, the Mistral model wins this by a sizable margin, with the second (Gemma) and third (Qwen) model being pretty close to each other.

Model choice

In the end, the Gemma model was chosen instead of the Mistral model. The Qwen model was disqualified by placing last in both metrics. The Gemma model was chosen due to several reasons:

1. Larger model: The Gemma model is more than 4 times larger than the Mistral model. This reflects in the generated texts where this model is able to be semantically richer and generate longer reviews which are still in coherent Slovak.

2. Instruct model: This makes tweaking the synthetic data to match the real data much easier, since things like expected length or sentiment can be inserted into the prompt.

Initial results of generating data

The initial synthetic half of the dataset was generated using the Gemma 3 model. However, while the reviews initially seemed good enough, they were very easily detectable with a simple model in a bigger sample size. A simple SVM model using TF-IDF as features immediately reached almost 100% accuracy and F1-score on the review dataset. Reasons for this were mostly the model using different wording despite being given examples than the dataset, it's tendency to repeat itself/talk about the same topics. Another reason was that it's sentiment was visibly much more polarized than the original dataset. For example, the model used exclamation marks in almost every single review, which could serve as a shortcut for the classifier to immediately detect it. For this reason, the Gemma 3 model was additionally fine-tuned on the review dataset.

Fine-tuning Gemma 3

A fine-tuning version of the dataset was created for both it's variants. The process was done as follows:

1. Reviews which were labeled as outliers due to their length (in words) were removed. The range for the review lengths to keep was $< Q1, Q3 + 1.5 * IQR >$, where $Q1$ and $Q3$ are the first and third quartile of review lengths respectively, while IQR is $Q3 - Q1$. Usually the left side of the non-outlier set is $Q1 - 1.5 * IQR$ [14], however since that went into the negative for this dataset, and the start of $Q1$ were reasonably long reviews for both variants, $Q1$ was used as the left bound.
2. For each remaining review, a instruct-response pair was created. The response was the sampled review, while the instruction was the original prompt for Gemma 3 with the review length/sentiment and a few tweaks, such as the rating label instructions being less polarized in sentiment (pozitívna, skôr pozitívna, skôr negatívna, negatívna).
3. The model was trained on the fine-tuning dataset and saved. The training parameters used are in Appendix 1. **[[do this]]**

4.5 Splitting the data for training and testing

Good splits of the data into training, validation and test splits are important to avoid problems during model training such as overfitting. They depend on the size of the dataset, and how balanced it is among other factors. A common ratio is splitting it into 80% training, and 10% per the validation and test splits [4]. Another common method is to use *K-Fold cross validation* [4], which splits the dataset into K folds. Then for each of these folds (so K times), it trains the model on the other $K - 1$ folds and validates it on the current fold. This ensures we use all of the data for both training and validation. Per the cited article, there are multiple techniques to choose „which data goes into each set“. If the dataset is balanced, meaning there isn't one group of the data larger than the other random sampling should suffice, however if the sets aren't equal, stratified sampling might be better. That means that the percentage of the data in each set for a given group corresponds to the

total percentage of the group in the dataset. The dataset is balanced when it comes to the volume of real and synthetic data, however it's not balanced when it comes to sentiment. For this reason, the data is sampled using the stratified strategy. As mentioned in [section 6.4](#), the chosen rating/content type was stored alongside the generated review. Then, when creating the training, validation and test splits, they were used as variable for their variant respectively alongside the class label (whether the review is real or synthetic) for stratified sampling. Concerning the ratio of the splits, multiple splits were created. A testing set containing 10% of the reviews was always created, and then the remaining 90% was split into 80-10 for training and validation in one split, and into 5 folds on the other split.

Chapter 5

Architecture of the classifier

For the classifier, a stacked ensemble model was chosen with 3 inner models. Each of the models focuses on different features of the dataset, and their outputs are fed as inputs to the meta learner, who makes the final prediction. [Figure 2](#) displays a high level overview of the proposed architecture.

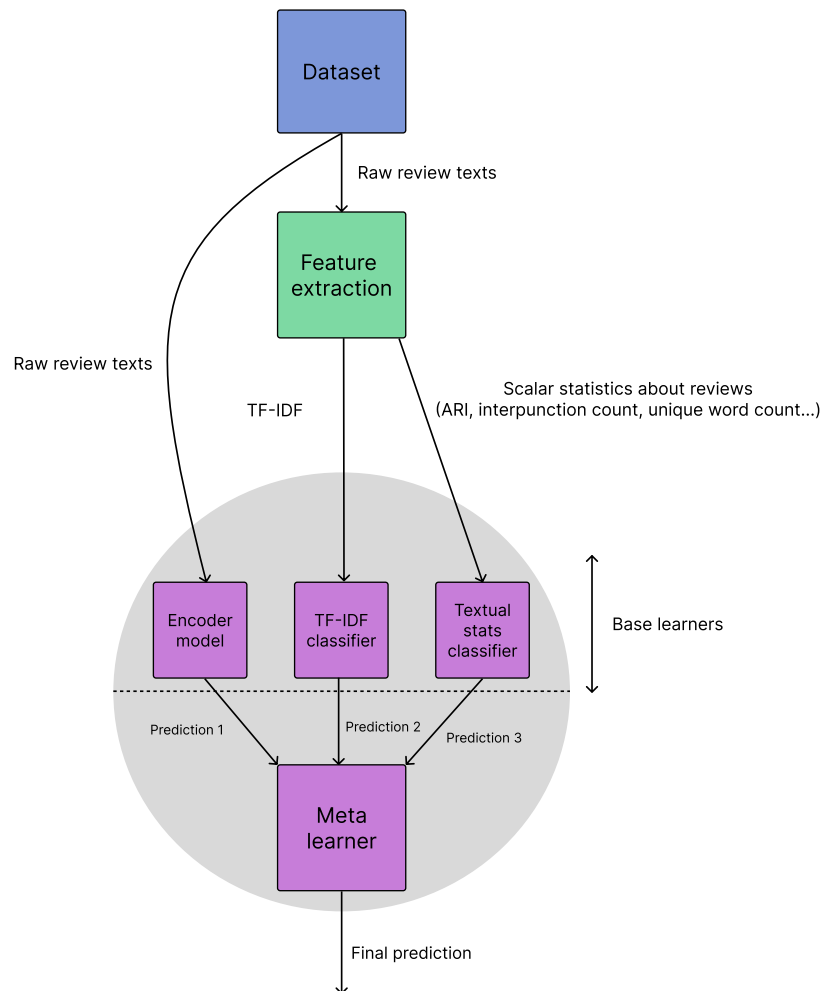


Figure 7: Architecture of the fake review detector.

The base learners are trained on either the raw review dataset, or vectors of features which are extracted from it to be used as input to the model. In the first stage, the base learners create a *meta dataset*, which contains probabilities of the review being fake for each model. Then the meta learner is trained on these probabilities, and it itself outputs a probability of the review being fake.

5.1 Model components

Encoder language model

One of the base learners is an encoder language model fine-tuned for review classification. The purpose of this component is to find semantic giveaways and patterns in the text which could indicate the text being artificially generated. Since this is a language model, it is trained on the raw review texts and the class labels, (which are internally turned into embeddings and then contextualized) and takes the raw text as input during inference.

TF-IDF classifier

Another base learner is a classifier which tries to predict the authenticity of the reviews upon their TF-IDF score. This is split from the textual stats classifier since the TF-IDF vector will be much larger than the vector of all other features, which are scalar values, as not to „drown out“ the other features.

Textual features classifier

The final base learner is a model which tries to classify the review based on textual features which are scalar values. This includes features such as ARI, TTR, count of punctuation characters, average sentence length, and others. The full list of features is provided in the implementation chapter.

Meta learner

This is the part of the model which actually decides the authenticity of the review. It takes in the predictions of the base models and outputs the probability of the review being AI generated.

Chapter 6

Implementation of the classifier

6.1 The classification framework

The classifier is implemented as a Python library. It includes options to use several models on different parts of the ensemble, choose different training procedures and turn on and off different parts of the ensemble. It utilizes the *scikit-learn*¹ library for most models, *PyTorch*² for the MLP model and the *Hugging Face Transformers*³ library for the Encoder model. The library has its core modules written as object-oriented and allows for easily adding another model, its configuration, or feature type.

6.2 Ensemble training algorithms

The library supports several algorithms for training the ensemble. The first, and the most simple one, trains the base models on the training set, runs inference on the same data and then trains the meta model on those predictions. However, this can cause the base models to be overly accurate and overfit. The second algorithm uses a validation set, where it trains the base models on the training set, runs inference on the validation set and then trains the meta model on the inferred predictions. However, this has the disadvantage that it only uses the validation set predictions to train the meta model, which is only 10% of the entire dataset. The algorithm used for training the ensemble is based on [8]. It uses *K-Fold cross validation* with $K = 5$ to train the base models. For each of the K training runs, it trains the base models on $K - 1$ folds. Then it runs inference of the base models on the remaining fold. The resulting predictions are used as $\frac{1}{K}$ of the resulting meta training set. This is done until inference is run on all 5 folds, and the model predictions compose the training dataset. The last step is retraining the base models on the entire training set. This causes both the base and meta models to use 100% of the training data. Figure Z showcases this on a high level:

6.3 Used models

The library includes several available models, where each of them is configurable as per its most usual parameters. Each model also includes the part of the ensemble that the model

¹<https://scikit-learn.org/stable/>

²<https://pytorch.org/>

³<https://huggingface.co/docs/transformers/index>

can be used for. Table N shows which models are available in the library and in which part of the ensemble they can be used:

Model	Statistical	TF-IDF	Transformer	Meta
SVM	Yes	Yes	No	Yes
KNN	Yes	No	No	No
Multinomial Naive Bayes	No	Yes	No	No
Logistic Regression	Yes	Yes	No	Yes
Random Forest	Yes	No	No	Yes
PyTorch MLP	Yes	No	No	No
Hugging Face Transformer	No	No	Yes	No

Table 7: Models supported by the library and their supported features.

All models are configured to output probabilities instead of just binary class labels, 0.5 is used as the threshold for a review being fake. To some models this comes naturally, such as Logistic Regression or an MLP, but in the case of an SVM, the scikit-learn API does this by training an additional Logistic Regression model on the SVMs scores and fitting it with additional cross validation on the training data ⁴.

6.4 Choosing optimal model parameters

To choose the optimal parameters for each model, the following was done:

1. Define parameter values to try out.
2. Create all possible combinations of those parameters for a given model.
3. Evaluate the model with the given parameters.
4. Choose the best performing combination.

This was done separately for both variants of the dataset. It was also done separately for all features the model supported, which means the optimal parameters could be different based on how the model was run, and with what data. For scikit-learn models, this was done using the `GridSearchCV` API ⁵. For Transformers models, it was done using Hugging Face’s `hyperparameter_search` ⁶ and for the PyTorch MLP it was done using a custom training loop which performed 5-fold cross-validation to evaluate.

6.5 The statistical classifier

This part of the ensemble uses the `StatisticalFeatureExtractor` class, which runs each feature extraction in parallel using Python’s `multiprocessing` module to create a process pool. The result is a dense feature matrix of size $(n_{samples}, n_{features})$ which is fed to the model. All input features are standardized using scikit-learn’s `StandardScaler` ⁷, which

⁴<https://scikit-learn.org/stable/modules/svm.html>

⁵https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html

⁶https://huggingface.co/docs/transformers/hpo_train

⁷<https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>

scales the data so that it's mean is 0, except for the Multinomial Naive Bayes, which expects counts and Random Forest, where the feature scale doesn't make a difference. The classifier uses the following features:

- **Focusing on length:** Sentence, word, and character count, mean word length and mean sentence length.
- **Focusing on punctuation:** The ratios of dots, commas, exclamation/question marks, colons and hyphens to total words.
- **Focusing on character case:** Ratios of all uppercase and title case words to other words.
- **Focusing on readability:** ARI.
- **Focusing on lexical diversity:** TTR, count of unique words, and Hapax-Legomenon rate.
- **Other:** Ratio of stop words to total words. The list of Slovak stop words is publicly available on GitHub.⁸ The reviews are stripped of diacritics before checking for stop words to avoid missing any.

6.6 The TF-IDF classifier

6.7 The Transformers classifier

⁸<https://github.com/stopwords-iso/stopwords-sk>

Chapter 7

Evaluation

7.1 Evaluation process

7.2 Results

Feature importance

7.3 Discussion and practical applications

Chapter 8

Conclusion

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